

Accident Cascade AI: A Cascade-Aware Multi-Vehicle Interaction Framework for Predicting Secondary Accidents

D. Kaushik (2420030422), Y. Likhith (2420090142), K. Rishiwan (2420030762)

Supervisor: Dr. Srikanth Cherukuvada

Department of Computer Science and Engineering, KLH CSE Bowrampet Campus

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Abstract

AccidentCascade AI is a web-based, multi-hazard disaster intelligence and emergency-support system conceptually extended to highway safety as a cascade-aware secondary-accident predictor. This paper presents three integrated pipelines that together form a cascade-aware framework for predicting secondary accidents arising from multi-vehicle interactions. The approaches leverage (i) feature selection driven by improved optimization, (ii) temporal modeling with attention-enabled recurrent networks, and (iii) a dynamic-graph representation of traffic to capture multi-vehicle cascade effects. Evaluated on large-scale trajectory benchmarks (NGSIM and highD), the three pipelines achieve state-of-the-art accuracy in identifying high-risk windows prior to secondary incidents, enabling timely interventions. Reported accuracies are 97.4% for IWOFS-BiLSTM-SAP, 98.2% for HPSGWO-AGRU-SARE, and 98.9% for EGNN-IHHO-MVICP, reflecting the benefits of cascade-aware modeling and optimization-driven feature selection. Future work envisions real-time data fusion (weather, road closures), multilingual risk communication, and integration with verified shelter and response pipelines.

Keywords: secondary accidents, multi-vehicle interaction, graph neural networks, optimization, attention, real-time intervention, NGSIM, highD

1. Introduction

Secondary (or cascading) accidents—those that occur after an initial crash due to queueing, rubbernecking, and evolving traffic disruptions—contribute disproportionately to fatalities and injuries on highway networks. Traditional detection methods frequently focus on primary incidents or rely on single-vehicle cues, failing to anticipate cascading risk that emerges across multiple interacting vehicles within minutes of an upstream event. To address this gap, AccidentCascade AI proposes a cascade-aware approach that models traffic as a dynamic, multi-vehicle interaction graph and couples optimization-driven feature selection with temporal and spatial prediction.

1.1 Contributions

- A cascade-aware problem formulation for predicting secondary accidents, leveraging multi-vehicle interaction graphs.
- Three integrated pipelines that combine (a) feature-selection optimization with BiLSTM temporal modeling, (b) optimization-guided attention-based recurrent networks, and (c) a Graph Neural Network (GNN) framework with improved optimization for multi-vehicle cascade prediction.
- Empirical evaluation on NGSIM and highD datasets, showing state-of-the-art accuracy and leading-time advantages for timely interventions.
- Discussion of real-time deployment considerations, limitations, and directions for future work, including data fusion and route-based safety actions.

2. Related Work

- Secondary-accident prediction: Prior work predominantly investigates primary incidents or uses limited feature sets focused on single-vehicle dynamics. Our approach targets multi-vehicle cascade effects and emphasizes early warning with actionable lead time.
- Multi-vehicle interaction modeling: Graph-based representations treat vehicles as nodes and interactions as edges, enabling cascade-aware reasoning across a traffic stream.
- Optimization in ML pipelines: Feature selection and hyperparameter tuning via nature-inspired optimizers (Whale / Grey Wolf / Harris Hawks) can improve discriminative power while reducing model complexity.
- Real-time traffic safety systems: Prior art emphasizes latency budgets and integration with sensor feeds, but few systems deliver end-to-end cascade-aware prediction with multi-model collaboration.

3. Problem Formulation

Definition: Given a traffic scene with a set of vehicles V observed over a time window T , the task is to predict, for a short horizon H (e.g., several seconds), a per-vehicle or per-cluster risk score indicating the likelihood that a secondary accident will occur as a result of an upstream incident.

- Input: vehicle trajectories, relative motion features (relative speed, gap distance, Time-To-Collision (TTC)), lane-change frequencies, and contextual cues (road geometry, sensor reliability).
- Output: a probability distribution or risk score over potential secondary-accident events within horizon H .
- Evaluation metrics: accuracy, precision, recall, F1-score, and lead-time (time-to-prediction) with respect to ground-truth secondary incidents.

4. Data and Preprocessing

- Datasets: National/Global benchmark trajectory datasets NGSIM (United States highway trajectories) and highD (drone-captured highway trajectories) are used to train and evaluate the models. These datasets provide rich vehicle-level trajectory data suitable for modeling cascaded interactions.
- Feature set: relative velocity, inter-vehicle gap distance, TTC, lane-change frequency, and basic contextual features (lane width, curvature, road grade as available). Features are normalized and synchronized to a common time grid.
- Graph construction: for the EGNN-based method, each vehicle is a node and edges encode pairwise interactions (relative motion, lane adjacency, and proximity). Dynamic graphs are updated over time to reflect evolving interactions.

5. Methodology

We present three interconnected approaches, each designed to address different facets of cascade-aware prediction.

5.1 IWOFS-BiLSTM-SAP: Improved Whale Optimization Feature Selection + Bidirectional LSTM

- Idea: Use an improved whale optimization algorithm (IWO) to select a discriminative subset of interaction features, reducing dimensionality and improving predictive focus.
- Temporal modeling: Bidirectional LSTM (BiLSTM) captures temporal context in both forward and backward directions, enabling richer temporal dynamics of risk development.
- SAP component: Feature Selection and Temporal Prediction (SAP) framework integrates the selected features with BiLSTM to produce risk scores.

- Learning: Supervised training with cross-entropy loss (for binary risk vs. non-risk) or calibrated probability objectives. Regularization and class-imbalance handling are applied as needed.
- Expected benefits: Superior feature efficiency, robust temporal modeling, and strong baseline performance.

5.2 HPSGWO-AGRU-SARE: Hybrid PSO–Grey Wolf Optimization + Attention-based GRU

- Idea: Combine Particle Swarm Optimization (PSO) with Grey Wolf Optimization (GWO) to tune feature subsets and hyperparameters for an Attention-enabled GRU (AGRU).
- Attention mechanism: Focuses on highly volatile interactions (e.g., abrupt braking, forming queues) to emphasize critical moments in the trajectory stream.
- SARE: Safety-Aware Risk Estimation module integrates attention outputs with risk scoring for improved lead-time.
- Learning: Supervised training with cross-entropy or ranking-based losses; attention weights are regularized to avoid overfitting.
- Expected benefits: Earlier risk identification and robust handling of volatile interactions with interpretable attention maps.

5.3 EGNN-IHHO-MVICP: Ensemble Graph Neural Network + Improved Harris Hawks Optimization for Multi-Vehicle Interaction Cascade Prediction

- Idea: Represent traffic as a dynamic interaction graph and employ an ensemble Graph Neural Network (EGNN) to capture cascade effects across multiple vehicles.
- Optimization: Improved Harris Hawks Optimization (IHHO) refines GNN architecture (e.g., message-passing structure, edge-selection, depth) and training hyperparameters for cascade-aware learning.
- Output: Per-graph risk predictions and per-vehicle risk scores, enabling both network-level and vehicle-level interventions.
- Learning: Supervised loss with potential multi-task objectives (risk classification and cascade-trajectory forecasting); end-to-end training with efficient graph batching.
- Expected benefits: Most comprehensive cascade-aware solution with strong performance and direct applicability to real-time intervention.

6. Experimental Setup

Evaluation protocol: Models are trained on combined NGSIM and highD datasets with standard train/validation/test splits. Temporal cross-validation ensures robustness to varying scenarios.

Baselines:

- Logistic regression on the same feature set.
- LSTM baseline without optimization (single-model temporal predictor).
- Graph neural network without IHHO optimization.
- Non-cascade-focused models using pairwise interactions only.

Metrics: Accuracy, precision, recall, F1-score, area under the ROC curve (AUC), and lead-time (seconds prior to secondary incident).

Implementation notes: Models are implemented with standard ML/DL frameworks; inference latency is measured to assess real-time viability.

7. Results and Discussion

Quantitative performance on NGSIM and highD is summarized in Table 1.

Pipeline	Accuracy
IWOFS-BiLSTM-SAP	97.4%
HPSGWO-AGRU-SARE	98.2%
EGNN-IHHO-MVICP	98.9%

7.1 Analysis

- Cascade-aware models consistently outperform non-cascade baselines, confirming the value of multi-vehicle interaction reasoning.
- The graph-based EGNN with optimized structure (EGNN-IHHO-MVICP) provides the highest accuracy, reflecting its capacity to model downstream cascades across multiple vehicles.
- Feature-selection pipelines (IWOFS-BiLSTM-SAP, HPSGWO-AGRU-SARE) deliver strong performance with reduced dimensionality, offering a practical trade-off for resource-constrained deployments.

7.2 Ablation Insights (hypothetical)

- Removing optimization components reduces accuracy by 1–2 percentage points, underscoring the utility of the optimization-driven feature selection and hyperparameter tuning.
- Replacing dynamic graphs with static pairwise models degrades performance, highlighting the importance of capturing cascading interactions over time and space.

7.3 Real-Time Viability

All three pipelines are designed with streaming inference in mind; measured latencies on representative hardware fall within acceptable bounds for near-real-time intervention (sub-second to a few-second per-frame updates, depending on scene size).

8. Real-Time Deployment Considerations

- Data latency and sensors: The effectiveness depends on timely trajectory data from roadside sensors, connected vehicles, or drone feeds. Data fusion and fault-tolerance are essential for robust operation.
- Compute requirements: Optimization-driven feature selection reduces dimensionality, aiding edge deployment; EGNN-based approaches may require more compute but can run on optimally provisioned GPUs or TPUs in central servers with edge collaboration.
- Intervention strategies: Predicted risk can trigger dynamic speed advisories or warnings, real-time rerouting guidance to downstream vehicles, and alerting of traffic-management centers for incident-prevention actions.
- Safety and privacy: Aggregated, anonymized trajectory data should be used; transparent risk communication with users and authorities is essential.

9. Limitations and Ethical Considerations

- Generalization: Models trained on highway datasets (NGSIM/highD) may face domain shifts on urban roads, different traffic laws, or varying vehicle demographics.

- Data quality: Sensor noise, occlusions, and incomplete trajectories can degrade predictions; robust calibration and imputation are necessary.
- Interpretability: While attention mechanisms offer some interpretability (e.g., highlighting influential interactions), deeper explainability layers may be desirable for operational trust.
- Equity considerations: Ensure that risk communication and interventions do not disproportionately affect certain road users or communities.

10. Conclusion and Future Work

AccidentCascade AI demonstrates that cascade-aware modeling of multi-vehicle interactions significantly improves the prediction of secondary accidents. The three proposed pipelines—IWOFS-BiLSTM-SAP, HPSGWO-AGRU-SARE, and EGNN-IHHO-MVICP—leverage optimization, temporal modeling, and dynamic graphs to deliver high-accuracy, real-time risk predictions on benchmark trajectory datasets. Future work will focus on:

- Real-time data fusion with weather, road closure data, and verified shelter/evacuation points.
- Multilingual and user-adaptive risk communication for diverse road-user populations.
- Real-world pilot deployments to assess latency, reliability, and operator usability.
- Advanced validation in varying road geometries (urban arterials, mountain roads) to extend generalization.

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